

MICROPROCESSOR & MICROCONTROLLER LABORATORY

SUBJECT CODE-PCC-EEE-692

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

LIST OF ASSIGNMENTS

Assignment No.	Details
Assignment 1	I. Write a program to find the sum of a series of 8-bit hexadecimal numbers. II. Write a program to perform the larger and smaller between two hexadecimal numbers.
Assignment 2	Write a program to perform multibyte addition and subtraction of two unsigned hexadecimal numbers.
Assignment 3	Write a 8086 program (with its respective machine code) to store a 16 bit data 9825H in AX. Then move the content of 16 bit data of AX to the memory locations 0301H and 0302H (Offset address). In continuation of above, store the memory address 0301H (Offset address) in BX. Then move the data from memory addressed by BX register to the CX register. What will be the content of CX register?
Assignment 4	25H and 98H are stored in 0301H and 0302H memory address respectively and .23H and 88H are stored in 0303H and 0304H memory address respectively. Now by a suitable 8086 program compare between the data 9825H and 8823H by CMP instruction. Show the status of Carry, Zero and Sign flags also.
Assignment 5	1. Let us see the following example:- <pre>MOV AX,VAR1 MOV BX,VAR2 CMP VAR1,VAR2 JCC LABEL MOV CX,VAR3 HLT LABEL:MOV DX,VAR4 HLT</pre> Here VAR1,VAR2,VAR3 and VAR 4 are four variable numbers. Explain the results with using these variable numbers (by choosing your own),when (i) VAR1>VAR2 and JCC=JZ,(ii)VAR1>VAR2 and JCC=JNZ,(iii)VAR1>VAR2 and JCC=JC,(iv)VAR1>VAR2 and

	JCC=JNC,(v)VAR1<VAR2 and JCC=JZ,(vi)VAR1<VAR2 and JCC=JNZ,(vii)VAR1<VAR2 and JCC=JC,(viii)VAR1<VAR2 and JCC=JNC,(ix)VAR1=VAR2 and JCC=JZ,(x)VAR1=VAR2 and JCC=JC
Assignment 6	I. What do you mean by 'LOOP' instruction in 8086 microprocessor. Explain by a suitable example. II. Write a program to insert five first odd numbers (i.e. 01H,03H,05H,07H AND 09H) in five consecutive physical address memory from 03000H to 03004H. III. Write a program to insert the first nine natural numbers (i.e. from 01H to 09H) in nine consecutive physical address memory from 05001H to 05009H.
Assignment 7	Write a program in assembly language of 8086 microprocessor to select largest 16 bit number in an 16 bit data array. The given data array is: 8341H,7258H,4639H,8453H AND 9630H. The counter should be stored in 03000H and 03001H physical addresses and data should be kept from 03002H to 0300BH physical addresses respectively.
Assignment 8	Perform a software simulation using Tinkercad to design an automatic traffic controller at a cross-section of two roads.
Assignment 9	Perform a software simulation using Tinkercad to design a theft alarm.
Assignment 10	Perform a software simulation to design a Light Sensor (Photoresistor) With Arduino in Tinkercad.